

Query profile

1 Загружаем данные

```
CREATE DATABASE IF NOT EXISTS task_15;
```

```
CREATE TABLE task_15.youtube
```

```
(
  `id` String,
  `fetch_date` DateTime,
  `upload_date_str` String,
  `upload_date` Date,
  `title` String,
  `uploader_id` String,
  `uploader` String,
  `uploader_sub_count` Int64,
  `is_age_limit` Bool,
  `view_count` Int64,
  `like_count` Int64,
  `dislike_count` Int64,
  `is_crawlable` Bool,
  `has_subtitles` Bool,
  `is_ads_enabled` Bool,
  `is_comments_enabled` Bool,
  `description` String,
  `rich_metadata` Array(Tuple(call String, content String, subtitle String, title String, url
String)),
  `super_titles` Array(Tuple(text String, url String)),
  `uploader_badges` String,
  `video_badges` String
)
ENGINE = MergeTree
ORDER BY (uploader, upload_date);;
```

```
INSERT INTO task_15.youtube
```

```
SELECT id,
  parseDateTimeBestEffortUS0rZero(toString(fetch_date)) AS fetch_date,
  upload_date AS upload_date_str,
  toDate(parseDateTimeBestEffortUS0rZero(upload_date::String)) AS upload_date,
  ifNull(title, '') AS title,
  uploader_id,
```

```
CREATE DATABASE IF NOT EXISTS task_15;
```

```
CREATE TABLE task_15.youtube
```

```
(
  `id` String,
  `fetch_date` DateTime,
  `upload_date_str` String,
  `upload_date` Date,
  `title` String,
  `uploader_id` String,
  `uploader` String,
  `uploader_sub_count` Int64,
  `is_age_limit` Bool,
  `view_count` Int64,
  `like_count` Int64,
  `dislike_count` Int64,
  `is_crawlable` Bool,
  `has_subtitles` Bool,
  `is_ads_enabled` Bool,
  `is_comments_enabled` Bool,
  `description` String,
```

```

    `rich_metadata` Array(Tuple(call String, content String, subtitle String, title String, url
String)),
    `super_titles` Array(Tuple(text String, url String)),
    `uploader_badges` String,
    `video_badges` String
)
ENGINE = MergeTree
    ORDER BY (uploader, upload_date);;

```

```

INSERT INTO task_15.youtube
SELECT id,
    parseDateTimeBestEffortUS0rZero(toString(fetch_date)) AS fetch_date,
    upload_date AS upload_date_str,
    toDate(parseDateTimeBestEffortUS0rZero(upload_date::String)) AS upload_date,
    ifNull(title, '') AS title,
    uploader_id,
    ifNull(uploader, '') AS uploader,
    uploader_sub_count,
    is_age_limit,
    view_count,
    like_count,
    dislike_count,
    is_crawlable,
    has_subtitles,
    is_ads_enabled,
    is_comments_enabled,
    ifNull(description, '') AS description,
    rich_metadata,
    super_titles,
    ifNull(uploader_badges, '') AS uploader_badges,
    ifNull(video_badges, '') AS video_badges
FROM s3('https://clickhouse-public-datasets.s3.amazonaws.com/youtube/original/files/*.zst',
    'JSONLines')
LIMIT 1000000
SETTINGS input_format_null_as_default = 1;

```

CREATE DATABASE IF NOT EXISTS task_15

Query id: abfb19c4-3e3c-4343-8ca1-a675d5c5a930

Ok.

0 rows in set. Elapsed: 0.014 sec.

```

CREATE TABLE task_15.youtube
(
    `id` String,
    `fetch_date` DateTime,
    `upload_date_str` String,
    `upload_date` Date,
    `title` String,
    `uploader_id` String,
    `uploader` String,
    `uploader_sub_count` Int64,
    `is_age_limit` Bool,
    `view_count` Int64,
    `like_count` Int64,
    `dislike_count` Int64,
    `is_crawlable` Bool,
    `has_subtitles` Bool,
    `is_ads_enabled` Bool,
    `is_comments_enabled` Bool,
    `description` String,
    `rich_metadata` Array(Tuple(
        call String,
        content String,
        subtitle String,
        title String,
        url String)),

```

```

    `super_titles` Array(Tuple(
        text String,
        url String)),
    `uploader_badges` String,
    `video_badges` String
)
ENGINE = MergeTree
ORDER BY (uploader, upload_date)

```

Query id: 04f479a0-2d4d-43f1-9fae-d614051aa718

Ok.

0 rows in set. Elapsed: 0.024 sec.

```

INSERT INTO task_15.youtube
SETTINGS input_format_null_as_default = 1
SELECT
    id,
    parseDateTimeBestEffortUS0rZero(toString(fetch_date)) AS fetch_date,
    upload_date AS upload_date_str,
    toDate(parseDateTimeBestEffortUS0rZero(CAST(upload_date, 'String')))) AS upload_date,
    ifNull(title, '') AS title,
    uploader_id,
    ifNull(uploader, '') AS uploader,
    uploader_sub_count,
    is_age_limit,
    view_count,
    like_count,
    dislike_count,
    is_crawtable,
    has_subtitles,
    is_ads_enabled,
    is_comments_enabled,
    ifNull(description, '') AS description,
    rich_metadata,
    super_titles,
    ifNull(uploader_badges, '') AS uploader_badges,
    ifNull(video_badges, '') AS video_badges
FROM s3('https://clickhouse-public-datasets.s3.amazonaws.com/youtube/original/files/*.zst',
'JSONLines')
LIMIT 1000000
SETTINGS input_format_null_as_default = 1

```

Query id: 05619f3b-9dd1-4cfb-8855-819f1ab12178

✓ Progress: 130.82 thousand rows, 34.66 MB (39.49 thousand rows/s., 10.46 MB/s.) ← Progress: 130.82 thousand rows, 34.66 MB (39.49 thousand rows/s., 10.46 MB/s.) ✎ Progress: 261.64 thousand rows, 69.35 MB (76.66 thousand rows/s., 20.32 MB/s.) ↑ Progress: 261.64 thousand rows, 69.35 MB (76.66 thousand rows/s., 20.32 MB/s.) ➤ Progress: 261.64 thousand rows, 69.35 MB (74.45 thousand rows/s., 19.73 MB/s.) → Progress: 261.64 thousand rows, 69.35 MB (74.4
 ✓ Progress: 1.15 million rows, 307.25 MB (110.88 thousand rows/s., 29.62 MB/s.) ← Progress: 1.15 million rows, 307.25 MB (109.82 thousand rows/s., 29.34 MB/s.) ✎ Progress: 1.15 million rows, 307.25 MB (108.89 thousand rows/s., 29.09 MB/s.) Ok.

0 rows in set. Elapsed: 10.562 sec. Processed 1.15 million rows, 307.25 MB (108.89 thousand rows/s., 29.09 MB/s.)
 Peak memory usage: 1.27 GiB.

2_ Выполняем два запроса. Строки, относящиеся к пробегу основного индекса в логах запросов выделены голубым в выводе лога.

```

SET send_logs_level = 'trace';

SELECT *
FROM task_15.youtube
WHERE like_count > 500
FORMAT Null;

```

```
SELECT *
FROM task_15.youtube
WHERE uploader = 'Various Artists - Topic'
FORMAT Null;
```

```
SET send_logs_level = 'trace'
```

Query id: 77a8c1e2-7653-4a17-8e4b-9114a7302284

Ok.

0 rows in set. Elapsed: 0.002 sec.

```
SELECT *
FROM task_15.youtube
WHERE like_count > 500
FORMAT `Null`
```

Query id: 0280620f-c9d1-4720-a555-2825dc73476f

```
[clickhouse-node] 2025.12.03 21:48:30.552767 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Debug> executeQuery: (from [::1]:48328) (query 2, line 3) SELECT * FROM task_15.youtube
WHERE like_count > 500 FORMAT Null; (stage: Complete)
[clickhouse-node] 2025.12.03 21:48:30.553528 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> Planner: Query to stage Complete
[clickhouse-node] 2025.12.03 21:48:30.553896 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> Planner: Query from stage FetchColumns to stage Complete
[clickhouse-node] 2025.12.03 21:48:30.554795 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> QueryPlanOptimizePrewhere: The min valid primary key position for moving to the
tail of PREWHERE is -1
[clickhouse-node] 2025.12.03 21:48:30.554845 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> QueryPlanOptimizePrewhere: Moved 1 conditions to PREWHERE
[clickhouse-node] 2025.12.03 21:48:30.555068 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Key
condition: unknown
[clickhouse-node] 2025.12.03 21:48:30.555148 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor):
Filtering marks by primary and secondary keys
[clickhouse-node] 2025.12.03 21:48:30.555622 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): PK index
has dropped 0/123 granules, it took 0ms across 3 threads.
[clickhouse-node] 2025.12.03 21:48:30.555757 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Selected
3/3 parts by partition key, 3 parts by primary key, 123/123 marks by primary key, 123 marks to read from 3 ranges
[clickhouse-node] 2025.12.03 21:48:30.555855 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor):
Spreading mark ranges among streams (default reading)
[clickhouse-node] 2025.12.03 21:48:30.556150 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Reading
approx. 1000000 rows with 4 streams
[clickhouse-node] 2025.12.03 21:48:30.556230 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> MergeTreeSelectProcessor: PREWHERE condition was split into 1 steps
[clickhouse-node] 2025.12.03 21:48:30.556317 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> MergeTreeSelectProcessor: PREWHERE condition was split into 1 steps
[clickhouse-node] 2025.12.03 21:48:30.556396 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> MergeTreeSelectProcessor: PREWHERE condition was split into 1 steps
[clickhouse-node] 2025.12.03 21:48:30.556473 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> MergeTreeSelectProcessor: PREWHERE condition was split into 1 steps
↑ Progress: 196.61 thousand rows, 152.79 MB (1.77 million rows/s., 1.38 GB/s.) ↗ Progress: 196.61 thousand rows, 152.79 MB (1.77 million rows/s., 1.38 GB/s.) → Progress: 491.52
thousand rows, 369.42 MB (2.32 million rows/s., 1.75 GB/s.) ↘ Progress: 491.52 thousand rows, 369.42 MB (2.32 million rows/s., 1.75 GB/s.) ↓ Progress: 761.09 thousand rows, 575.38
MB (2.43 million rows/s., 1.83 GB/s.) ✓ Progress: 761.09 thousand rows, 575.38 MB (2.43 million rows/s., 1.83 GB/s.) [clickhouse-node] 2025.12.03 21:48:30.866646 [ 4639 ]
{0280620f-c9d1-4720-a555-2825dc73476f} <Trace> MergeTreeSelectProcessor: Read steps statistics: step 0 rows_read: 196608, step 1 rows_read: 196608
[clickhouse-node] 2025.12.03 21:48:30.898396 [ 4675 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> MergeTreeSelectProcessor: Read steps statistics: step 0 rows_read: 215092, step 1
rows_read: 215092
[clickhouse-node] 2025.12.03 21:48:30.930346 [ 4644 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> MergeTreeSelectProcessor: Read steps statistics: step 0 rows_read: 196608, step 1
rows_read: 196608
↖ Progress: 910.65 thousand rows, 692.85 MB (2.18 million rows/s., 1.66 GB/s.) ↙ Progress: 910.65 thousand rows, 692.85 MB (2.18 million rows/s., 1.66 GB/s.) [clickhouse-node]
2025.12.03 21:48:31.074638 [ 4666 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Trace> MergeTreeSelectProcessor: Read steps statistics: step 0 rows_read: 391692, step 1 rows_read: 391692
[clickhouse-node] 2025.12.03 21:48:31.076525 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Debug> executeQuery: Read 1000000 rows, 722.36 MiB in 0.523749 sec., 1909311.5213585133
rows/sec., 1.35 GiB/sec.
[clickhouse-node] 2025.12.03 21:48:31.077425 [ 3977 ] {0280620f-c9d1-4720-a555-2825dc73476f} <Debug> MemoryTracker: Query peak memory usage: 89.55 MiB.
Ok.
```

0 rows in set. Elapsed: 0.524 sec. Processed 1.00 million rows, 757.45 MB (1.91 million rows/s., 1.45 GB/s.)
Peak memory usage: 89.55 MiB.

```
SELECT *
FROM task_15.youtube
WHERE uploader = 'Various Artists - Topic'
FORMAT `Null`
```

Query id: bbc13eab-ddf1-41e4-b453-a631c1fc3820

```
[clickhouse-node] 2025.12.03 21:48:31.079359 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Debug> executeQuery: (from [::1]:48328) (query 3, line 8) SELECT * FROM task_15.youtube
WHERE uploader = 'Various Artists - Topic' FORMAT Null; (stage: Complete)
[clickhouse-node] 2025.12.03 21:48:31.080194 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> Planner: Query to stage Complete
[clickhouse-node] 2025.12.03 21:48:31.080570 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> Planner: Query from stage FetchColumns to stage Complete
[clickhouse-node] 2025.12.03 21:48:31.081610 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> QueryPlanOptimizePrewhere: The min valid primary key position for moving to the
tail of PREWHERE is 0
[clickhouse-node] 2025.12.03 21:48:31.081658 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> QueryPlanOptimizePrewhere: Moved 1 conditions to PREWHERE
[clickhouse-node] 2025.12.03 21:48:31.081879 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Key
condition: (column 0 in ['Various Artists - Topic', 'Various Artists - Topic'])
[clickhouse-node] 2025.12.03 21:48:31.081928 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor):
Filtering marks by primary and secondary keys
[clickhouse-node] 2025.12.03 21:48:31.082238 [ 4656 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Running
binary search on index range for part all_1_1_0 (49 marks)
[clickhouse-node] 2025.12.03 21:48:31.082306 [ 4668 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Running
binary search on index range for part all_3_3_0 (28 marks)
[clickhouse-node] 2025.12.03 21:48:31.082364 [ 4668 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Found
(LEFT) boundary mark: 18
[clickhouse-node] 2025.12.03 21:48:31.082331 [ 4656 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Found
(LEFT) boundary mark: 34
[clickhouse-node] 2025.12.03 21:48:31.082310 [ 4635 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Running
binary search on index range for part all_2_2_0 (49 marks)
[clickhouse-node] 2025.12.03 21:48:31.082388 [ 4668 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Found
(RIGHT) boundary mark: 19
[clickhouse-node] 2025.12.03 21:48:31.082394 [ 4656 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Found
(RIGHT) boundary mark: 35
[clickhouse-node] 2025.12.03 21:48:31.082414 [ 4668 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Found
continuous range in 8 steps
```

```

[clickhouse-node] 2025.12.03 21:48:31.082421 [ 4656 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Found continuous range in 9 steps
[clickhouse-node] 2025.12.03 21:48:31.082430 [ 4635 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Found (LEFT) boundary mark: 34
[clickhouse-node] 2025.12.03 21:48:31.082527 [ 4635 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Found (RIGHT) boundary mark: 35
[clickhouse-node] 2025.12.03 21:48:31.082560 [ 4635 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Found continuous range in 9 steps
[clickhouse-node] 2025.12.03 21:48:31.082762 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): PK index has dropped 120/123 granules, it took 0ms across 3 threads.
[clickhouse-node] 2025.12.03 21:48:31.082824 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Selected 3/3 parts by partition key, 3 parts by primary key, 3/123 marks by primary key, 3 marks to read from 3 ranges
[clickhouse-node] 2025.12.03 21:48:31.082852 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Spreading mark ranges among streams (default reading)
[clickhouse-node] 2025.12.03 21:48:31.083078 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Reading approx. 24576 rows with 3 streams
[clickhouse-node] 2025.12.03 21:48:31.083120 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> MergeTreeSelectProcessor: PREWHERE condition was split into 1 steps
[clickhouse-node] 2025.12.03 21:48:31.083178 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> MergeTreeSelectProcessor: PREWHERE condition was split into 1 steps
[clickhouse-node] 2025.12.03 21:48:31.083229 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> MergeTreeSelectProcessor: PREWHERE condition was split into 1 steps
[clickhouse-node] 2025.12.03 21:48:31.104911 [ 4626 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> MergeTreeSelectProcessor: Read steps statistics: step 0 rows_read: 8192, step 1 rows_read: 3153
[clickhouse-node] 2025.12.03 21:48:31.106476 [ 4680 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> MergeTreeSelectProcessor: Read steps statistics: step 0 rows_read: 8192, step 1 rows_read: 8192
[clickhouse-node] 2025.12.03 21:48:31.107518 [ 4681 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Trace> MergeTreeSelectProcessor: Read steps statistics: step 0 rows_read: 8192, step 1 rows_read: 3321
Progress: 24.58 thousand rows, 11.29 MB (834.90 thousand rows/s., 383.58 MB/s.) Progress: 24.58 thousand rows, 11.29 MB (834.90 thousand rows/s., 383.58 MB/s.) [clickhouse-node] 2025.12.03 21:48:31.108648 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Debug> executeQuery: Read 24576 rows, 10.77 MiB in 0.029345 sec., 837485.0911569262 rows/sec., 366.94 MiB/sec.
[clickhouse-node] 2025.12.03 21:48:31.109401 [ 3977 ] {bbc13eab-ddf1-41e4-b453-a631c1fc3820} <Debug> MemoryTracker: Query peak memory usage: 31.95 MiB.
Ok.

```

0 rows in set. Elapsed: 0.030 sec. Processed 24.58 thousand rows, 11.29 MB (831.09 thousand rows/s., 381.83 MB/s.)
Peak memory usage: 31.95 MiB.

3_ Выводим с explain и показываем использование индекса

```

clickhouse-node.ru-central1.internal :) EXPLAIN indexes = 1
SELECT *
FROM task_15.youtube
WHERE like_count > 500;

```

```

EXPLAIN indexes = 1
SELECT *
FROM task_15.youtube
WHERE like_count > 500

```

Query id: 3d4e1966-5b71-470b-a9c6-e5d690479c9a

```

[clickhouse-node] 2025.12.03 21:54:06.927515 [ 3977 ] {3d4e1966-5b71-470b-a9c6-e5d690479c9a} <Debug> executeQuery: (from [::1]:48328) (query 1, line 1) EXPLAIN indexes = 1 SELECT * FROM task_15.youtube WHERE like_count > 500; (stage: Complete)
[clickhouse-node] 2025.12.03 21:54:06.928549 [ 3977 ] {3d4e1966-5b71-470b-a9c6-e5d690479c9a} <Trace> Planner: Query to stage Complete
[clickhouse-node] 2025.12.03 21:54:06.929273 [ 3977 ] {3d4e1966-5b71-470b-a9c6-e5d690479c9a} <Trace> Planner: Query from stage
FetchColumns to stage Complete
[clickhouse-node] 2025.12.03 21:54:06.930592 [ 3977 ] {3d4e1966-5b71-470b-a9c6-e5d690479c9a} <Trace> QueryPlanOptimizePrewrite: The min valid primary key position for moving to the tail of PREWHERE is -1
[clickhouse-node] 2025.12.03 21:54:06.930687 [ 3977 ] {3d4e1966-5b71-470b-a9c6-e5d690479c9a} <Trace> QueryPlanOptimizePrewrite: Moved 1 conditions to PREWHERE
[clickhouse-node] 2025.12.03 21:54:06.931013 [ 3977 ] {3d4e1966-5b71-470b-a9c6-e5d690479c9a} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Key condition: unknown
[clickhouse-node] 2025.12.03 21:54:06.931064 [ 3977 ] {3d4e1966-5b71-470b-a9c6-e5d690479c9a} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): Filtering marks by primary and secondary keys
[clickhouse-node] 2025.12.03 21:54:06.932462 [ 3977 ] {3d4e1966-5b71-470b-a9c6-e5d690479c9a} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423) (SelectExecutor): PK index has dropped 0/123 granules, it took 0ms across 3 threads.
[clickhouse-node] 2025.12.03 21:54:06.934074 [ 3977 ] {3d4e1966-5b71-470b-a9c6-e5d690479c9a} <Debug> executeQuery: Read 8 rows, 258.00 B in 0.006565 sec., 1218.5833968012187 rows/sec., 38.38 KiB/sec.

```

```

explain
1. Expression ((Project names + Projection))
2. Expression
3. ReadFromMergeTree (task_15.youtube)
4. Indexes:
5. PrimaryKey
6. Condition: true
7. Parts: 3/3
8. Granules: 123/123

```

8 rows in set. Elapsed: 0.007 sec.

```

EXPLAIN indexes = 1
SELECT *
FROM task_15.youtube
WHERE uploader = 'Various Artists - Topic';

```

```

EXPLAIN indexes = 1
SELECT *
FROM task_15.youtube

```


WHERE uploader = 'Various Artists - Topic'

Query id: 05c629d2-a275-451e-b82b-edf494ac5fbf

```
[clickhouse-node] 2025.12.03 21:55:03.338450 [ 3977 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Debug> executeQuery: (from [::1]:48328) (query 1, line 1)
EXPLAIN indexes = 1 SELECT * FROM task_15.youtube WHERE uploader = 'Various Artists - Topic'; (stage: Complete)
[clickhouse-node] 2025.12.03 21:55:03.358452 [ 3977 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> Planner: Query to stage Complete
[clickhouse-node] 2025.12.03 21:55:03.358984 [ 3977 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> Planner: Query from stage FetchColumns to stage
Complete
[clickhouse-node] 2025.12.03 21:55:03.360207 [ 3977 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> QueryPlanOptimizePrewhere: The min valid primary key
position for moving to the tail of PREWHERE is 0
[clickhouse-node] 2025.12.03 21:55:03.360267 [ 3977 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> QueryPlanOptimizePrewhere: Moved 1 conditions to
PREWHERE
[clickhouse-node] 2025.12.03 21:55:03.360517 [ 3977 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Key condition: (column 0 in ['Various Artists - Topic', 'Various Artists - Topic'])
[clickhouse-node] 2025.12.03 21:55:03.360574 [ 3977 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Filtering marks by primary and secondary keys
[clickhouse-node] 2025.12.03 21:55:03.360932 [ 4630 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Running binary search on index range for part all_1_1_0 (49 marks)
[clickhouse-node] 2025.12.03 21:55:03.361012 [ 4630 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Found (LEFT) boundary mark: 34
[clickhouse-node] 2025.12.03 21:55:03.361047 [ 4630 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Found (RIGHT) boundary mark: 35
[clickhouse-node] 2025.12.03 21:55:03.361078 [ 4630 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Found continuous range in 9 steps
[clickhouse-node] 2025.12.03 21:55:03.364005 [ 4630 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Running binary search on index range for part all_2_2_0 (49 marks)
[clickhouse-node] 2025.12.03 21:55:03.364085 [ 4630 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Found (LEFT) boundary mark: 34
[clickhouse-node] 2025.12.03 21:55:03.364126 [ 4630 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Found (RIGHT) boundary mark: 35
[clickhouse-node] 2025.12.03 21:55:03.364154 [ 4630 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Found continuous range in 9 steps
[clickhouse-node] 2025.12.03 21:55:03.364149 [ 4623 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Running binary search on index range for part all_3_3_0 (28 marks)
[clickhouse-node] 2025.12.03 21:55:03.364274 [ 4623 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Found (LEFT) boundary mark: 18
[clickhouse-node] 2025.12.03 21:55:03.364319 [ 4623 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Found (RIGHT) boundary mark: 19
[clickhouse-node] 2025.12.03 21:55:03.364352 [ 4623 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Trace> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): Found continuous range in 8 steps
[clickhouse-node] 2025.12.03 21:55:03.364580 [ 3977 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Debug> task_15.youtube (b5592f68-5202-49d1-93dc-0c187bd8a423)
(SelectExecutor): PK index has dropped 120/123 granules, it took 0ms across 3 threads.
[clickhouse-node] 2025.12.03 21:55:03.366305 [ 3977 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Debug> executeQuery: Read 10 rows, 370.00 B in 0.028012 sec.,
356.98986148793375 rows/sec., 12.90 KiB/sec.
[clickhouse-node] 2025.12.03 21:55:03.366651 [ 3977 ] {05c629d2-a275-451e-b82b-edf494ac5fbf} <Debug> MemoryTracker: Query peak memory usage: 430.19 KiB.
```

explain

1. Expression ((Project names + Projection))

2. Expression

3. ReadFromMergeTree (task_15.youtube)

4. Indexes:

5. PrimaryKey

6. Keys: _

7. uploader

8. Condition: (uploader in ['Various Artists - Topic', 'Various Artists - Topic'])

9. Parts: 3/3

10. Granules: 3/123

10 rows in set. Elapsed: 0.029 sec.

Заголовок